

Extracellular matrix viscoelasticity regulates mammary branching morphogenesis

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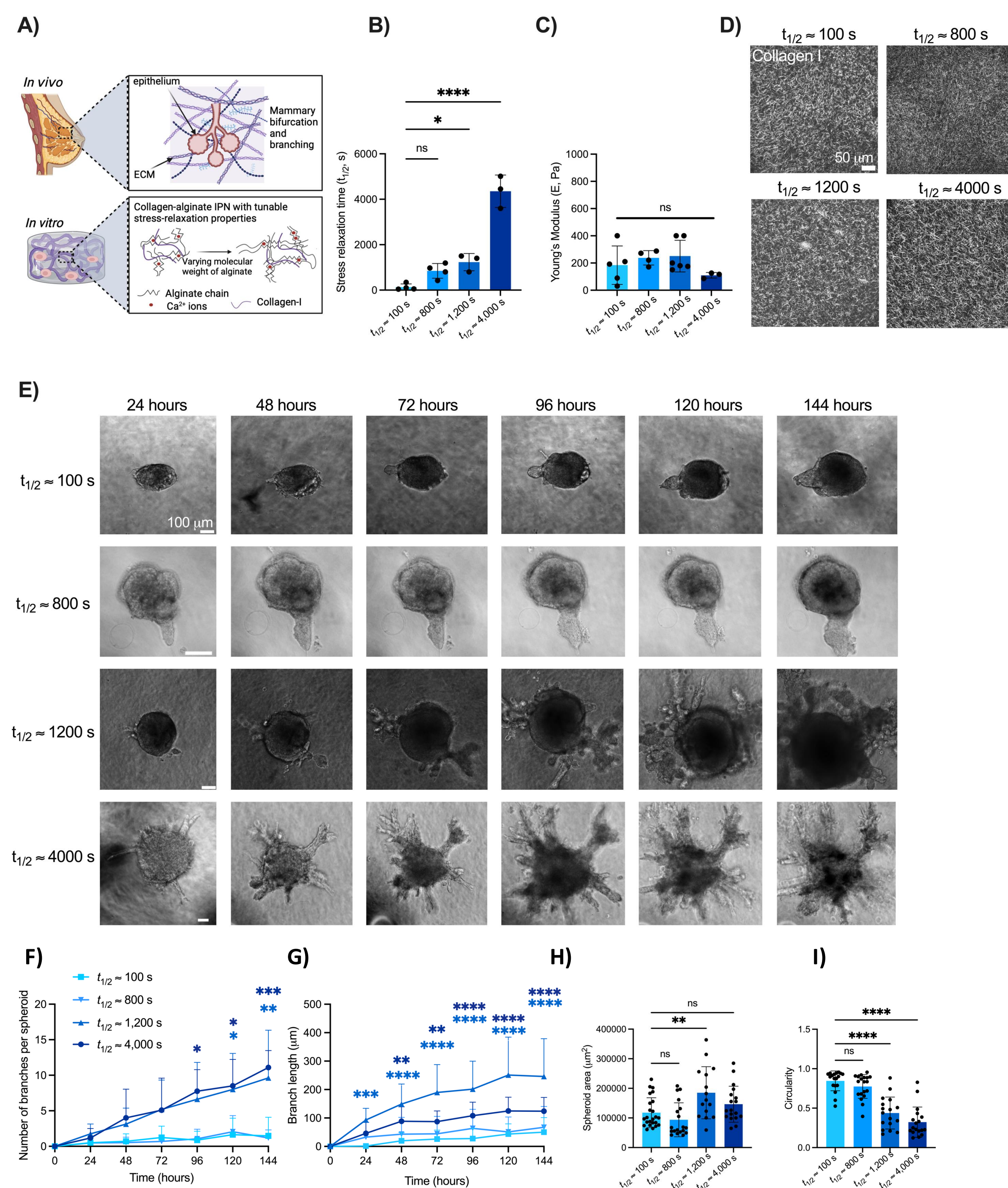
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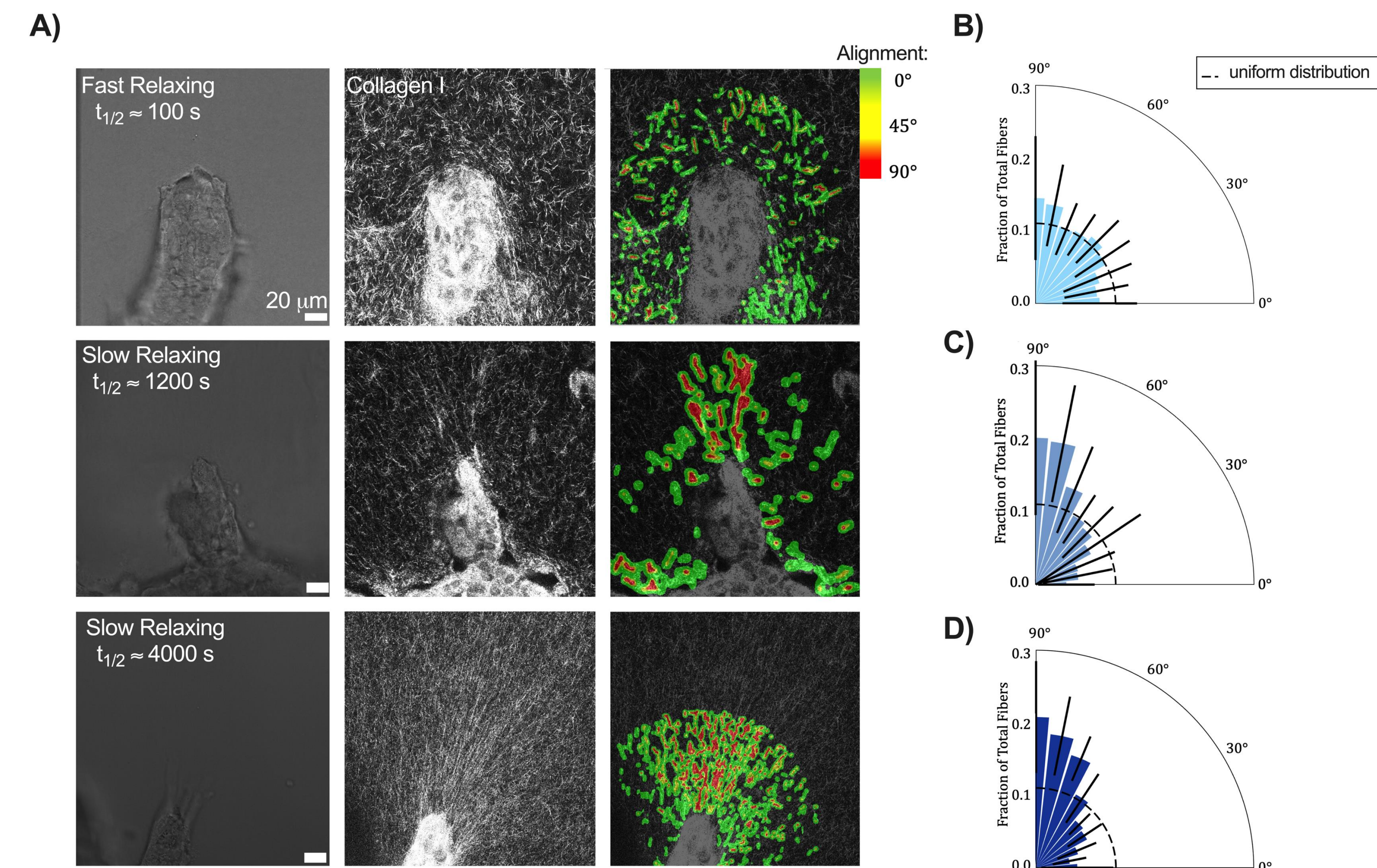
The effect of viscoelasticity on tissue organization is unknown

Branching morphogenesis is a fundamental process that occurs during the development of mammary tissue, and several signaling pathways that regulate mammary branching are disrupted during the progression of breast cancer. Mammary gland development is influenced by mechanical and chemical cues within its *in vivo* microenvironment, and conventionally, tissue development has been studied with *in vivo* or *ex vivo/in vitro* systems where mechanical properties are poorly controlled in space and time, lack tunability, or do not mimic the native extracellular matrix (ECM) mechanics. Here, we engineered a collagen-alginate interpenetrating network (IPN) to recapitulate the mammary *in vivo* microenvironment and determine how ECM viscoelasticity regulates tissue branching in mammary development.

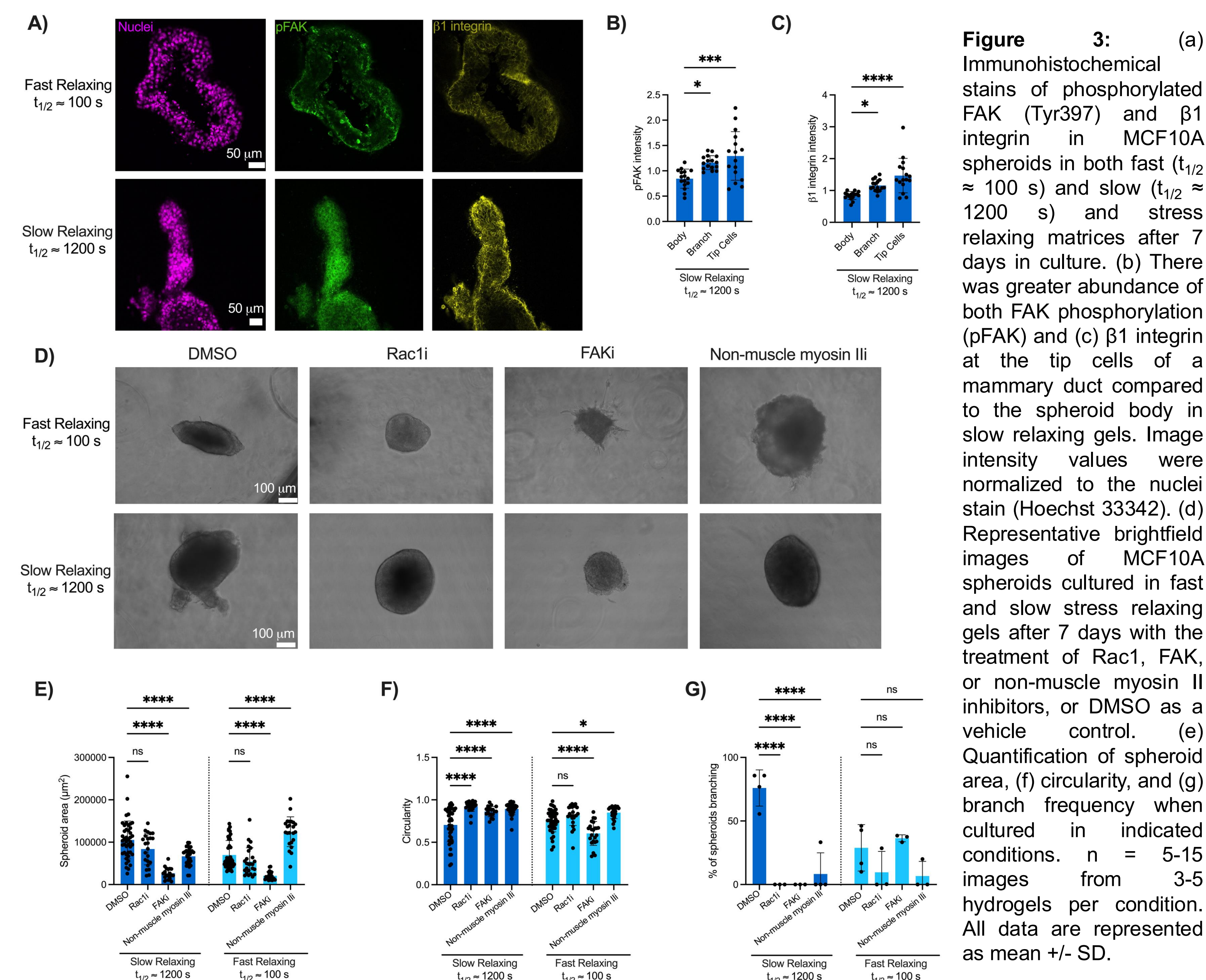
Mammary epithelial spheroid morphology and branch formation varies with stress-relaxation



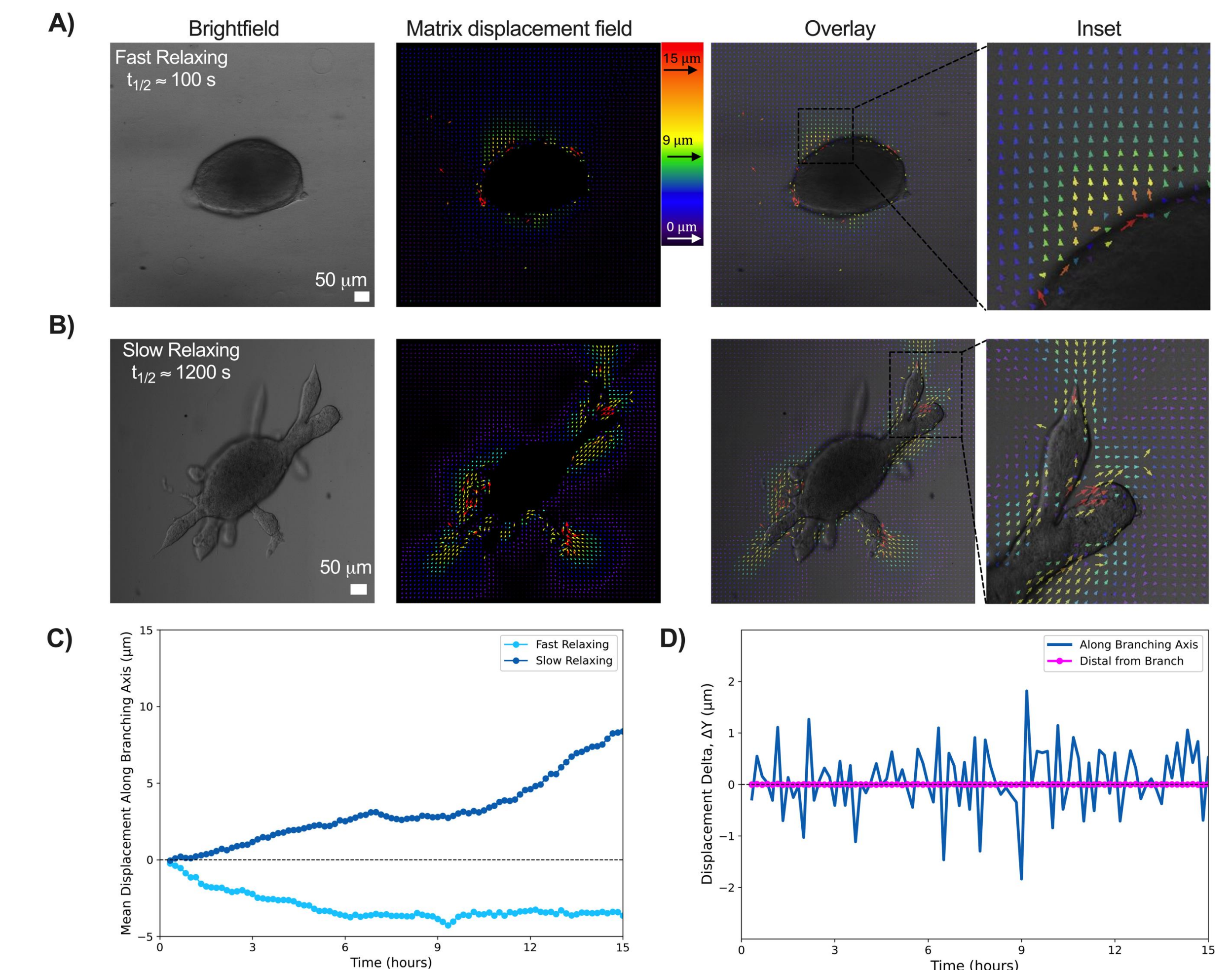
Slow stress-relaxing matrices promote collagen fiber alignment along branching axis



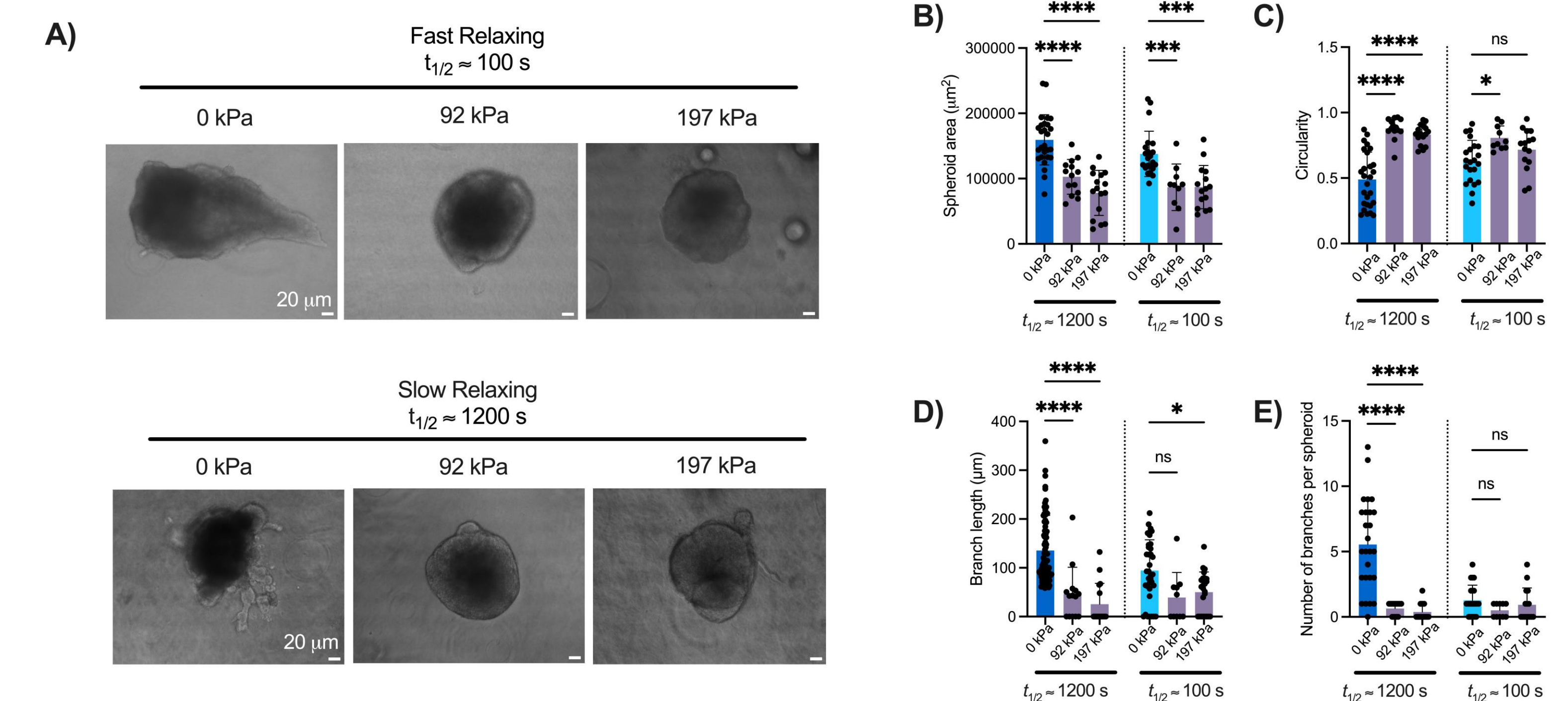
Actomyosin contractility and phosphorylated focal adhesion kinase contribute to mammary branching



Cell-generated non-continuous contractions displace the matrix to enable branching in slow stress-relaxing gels



Hyperosmotic stress modulates mammary growth in slow stress-relaxing matrices



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